

Dr. Yonatan Ashenafi

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Experience

Postdoctoral Fellow, Worcester Polytechnic Institute (WPI)

Aug 2023 -

Worcester, MA

- Conducting research on modeling intracellular transport processes using experimental data from multi-university collaborations.
- Teaching Multivariable Calculus, Calculus I & II, and Ordinary Differential Equations.

Postdoctoral Fellow, University of Alberta

Sep 2021 – May 2023

Edmonton, AB

- Researched motion mechanisms in diatoms (microscopic algae of ecological importance) through video-based data analysis.
- Taught Calculus I for Life Sciences and Calculus II for Physical Sciences.

Technical Fellow Intern, General Electric (GE) Research

Oct – Dec 2020

GE Global Research Center, Niskayuna, NY

- Collaborated with the Probabilistic Design Team on experimental design and additive manufacturing projects.
- Developed Python programs for reinforcement learning tasks.

Research Assistant, Dordt College

Jun – Aug 2014

Sioux Center, IA

- Constructed Gaussian mixture models from bacterial gene expression data using R.
- Gained experience in statistical data analysis and biological data modeling.

Teaching Assistant, Department of Mathematical Sciences

2016 – 2021

Rensselaer Polytechnic Institute (RPI), Troy, NY

- Led recitation sections, office hours, and exam review sessions for undergraduate courses.
- Assisted with instruction, grading, and student support for courses including Calculus I–III, Introduction to Differential Equations, Probability, Introduction to Data Mathematics, and Foundations of Applied Mathematics.

Tutor, Dordt College Library

Sep 2013 – May 2016

Sioux Center, IA

- Tutored students in College Algebra, Calculus I–III, Linear Algebra, and Discrete Mathematics.
- Collaborated effectively with students, faculty, and staff to improve academic performance.

Education

Rensselaer Polytechnic Institute (RPI), Troy, NY

Ph.D. in Mathematics, May 2021

Thesis: *Stochastic Hydrodynamics of Colonial Microswimmers*
M.S. in Applied Mathematics, August 2016 – December 2018

Dordt University, Sioux Center, IA

B.A. in Mathematics, August 2012 – May 2016

Publications

- Ashenafi, Y. & Kramer, P. R. (2024). Statistical Mobility of Multicellular Colonies of Flagellated Swimming Cells. *Bulletin of Mathematical Biology*.
- Ashenafi, Y. (2022). Stability and Spatial Autocorrelations of Suspensions of Microswimmers with Heterogeneous Spin. *Physics of Fluids*.
- Ashenafi, Y., Pandita, P., & Ghosh, S. (2021). Reinforcement Learning-Based Sequential Batch-Sampling for Bayesian Optimal Experimental Design. *Journal of Mechanical Design*.
- Zheng, P. et al. (2023). Cooperative Motility, Force Generation and Mechanosensing in a Foraging Non-Photosynthetic Diatom. *Open Biology*, 13(10), 230148.
- Disselkoen, C., et al. (2016). A Bayesian Framework for the Classification of Microbial Gene Activity States. *Frontiers in Microbiology*, 7, 1191.
- Ashenafi, Y. & Kramer, P. R. Stochastic Analysis of Kinesis and Taxis for Colonial Protozoa. Under review.
- Ashenafi, Y. Linearizing Nonlinear Stochastic Dynamics via Bijective State-Space Transformations for Bayesian Tracking. In preparation.
- Ashenafi, Y. & Walcott, S. Simulation-Based Inference of Hidden Motor States in Intracellular Cargo Transport. In preparation.

Service and Activities

- Proposal Reviewer, National Science Foundation (NSF), 2025–present.
- Presenter, 2025 Canadian Mathematical Society Mathematics Education Meeting (Online).
- Judge, SIMIODE SCUDEM X 2025 (Differential Equations Modeling Challenge).
- Volunteer Grader, WPI Annual Math Meet, 2024.
- Presenter, Ethiopian Mathematics Professional Association Seminar, July 2025.
- Treasurer, Black Graduate Students Association, RPI (2020). Managed finances during the COVID-19 pandemic.
- Organizer, Mini-Symposium on Noise and Asymmetry in Microscopic Life, CAIMS 2022.
- Mathematical Modeler, Mathematical Problems in Industry (MPI) Workshop, Univ. of Vermont (2024) and RPI (2017).
- Participant, SIMIODE Spring 2024 Webinars.
- Presenter at Biophysical Society (2025),

- Presenter at Biology and Medicine through Mathematics (BAMM) conference (2024),
- Attended the Institute for Computational and Experimental Research in Mathematics (ICERM) Workshop on Interacting Particle Systems: Analysis, Control, Learning, and Computation (2024),
- Presenter at Society for Industrial and Applied Mathematics (SIAM) Life Sciences conference (2021),
- Presenter at Mathematical Biosciences Institute Workshop (2020),
- Presenter at Univ. of Minnesota Multiscale Modeling (2019),
- Judge, University of Alberta Undergraduate Research Symposium, 2022.

Awards

- Bill and Nancy Siegmann Applied Mathematical Modeling Prize, RPI (2021).
- DiPrima Summer Research Fellowship, RPI (2018).
- ASME CIE Advanced Modeling and Simulation Best Paper Award (2022).
- Viss Mathematics Scholarship (2016)

Technical Skills

Python (NumPy, Pandas, SciPy, TensorFlow, scikit-learn), MATLAB, C++, R, SQL

Professional Membership

Society for Industrial and Applied Mathematics (SIAM)